# BANGALORE INSTITUTE OF TECHNOLOGY

An Autonomous Institution under Visvesvaraya Technological University, Belagavi
K. R. Road, V. V. Puram, Bengaluru - 560004

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING


## MODULE-WISE QUESTION BANK

| | | | |
|---|---|---|---|
| **Academic Year :** | **2026-2027** | **Batch :** | **2024-2028** |
| **Semester & Section :** | **V** | **Course Title & Course Code :** | **Theory of Computation (BCS503)** |
| **Credits :** | **4** | **Course Coordinator :** | **Prof Lakshmi B N** |
| **Total Contact Hours :** | **4** | **Total Hours of Pedagogy:** | |
| **Faculty In-charge(s) :** | **Prof. Chaithra G V** | **No. of Modules :** | **1** |

**Guidelines:**

- Questions shall be mapped to the appropriate Course Outcome (CO).
- Bloom's cognitive level shall be indicated as L1 to L6, wherever applicable.
- Question category may be indicated as Conceptual / Numerical / Application / Analysis / Design / Case Study / Programming / Other, as applicable.
- The question bank shall provide adequate coverage of all modules and relevant COs.

**Bloom's Taxonomy Reference**

| L1 | L2 | L3 | L4 | L5 | L6 |
|---|---|---|---|---|---|
| Remember | Understand | Apply | Analyse | Evaluate | Create |

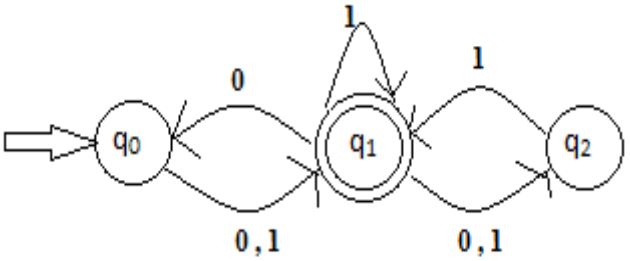
### MODULE - 1: Introduction to Automata Theory

| Sl. No. | MODULE – 1 | CO's | Blooms Levels |
|---|---|---|---|
| 1 | Define the following with suitable example<br>a. Alphabet<br>b. Power of an alphabet<br>c. Symbols<br>d. Strings<br>e. Languages<br>f. DFA<br>g. NFA | 1 | 1 |
| 2 | Define Automata Theory? Explain why we need to study automata theory | 1 | 2 |
| 3 | Explain with an example when we say that a Language is accepted by a Machine? | 1 | 2 |
| 4 | Explain the applications of Automata Theory. | 1 | 2 |
| 5 | With a neat diagram explain the Hierarchy of Language Classes. | 1 | 2 |
| 6 | Outline DFA ,NFA & ε -NFA with an example. | 1 | 2 |
| 7 | Contrast between DFA,NFA & ε NFA | 1 | 2 |
| 8 | Build DFA to accept the following language over the alphabet { 0 , 1 }<br>(i) L = { w / w is a even number}<br>(ii) L = { $(0\ 1)^i\ 1^{2j}$ \| i>= 1, j >= 1 }<br>(iii) The set of strings either starts or ends or both with 01<br>(iv) Set of all strings starting with at least two 0's and ending with at least two 1's<br>(v) The strings with consecutive 0's<br>(vi) Set of all strings not containing 110<br>(vii) Set of all string with exactly three consecutive 0's | 2 | 3 |

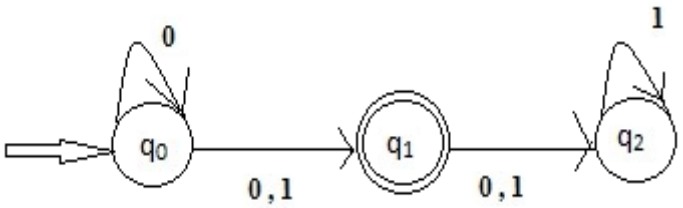
| | | | |
|---|---|---|---|
| | (viii) Set of all string having substring 001<br>(ix) Set of all strings such that number of 1's is even and the number of 0's is a multiple of<br>(x) L = { w \| w is of even length and begins with 01}<br>(xi) L = { w \| w consists of 0's and 1's with 011 as substring }<br>(xii) L = { w \| w ∈ 0 ( 1 )$^n$ 2; where n is an odd number} | | |
| 9 | Build DFA for the following L on ∑ = {a, b}<br><br>(i) The set of all strings that either begins or ends or both with substring ab<br>(ii) The set of all strings that ends with substring abb<br>(iii) The set of all strings not containing the substring aab<br>(iv) L = { w : \|w\| mod 5 <> 0} or L = { w : \|w\| mod 5 = 0}<br>(v) The set of all strings which do not ends with substring abb<br>(vi) L = { a w a \| w ∈ { a, b }$^*$ }<br>(vii) Set of all strings with at least one 'a' and exactly two b's<br>(viii) L = { w \| $n_a$ ( w ) <= 4; where w ∈ { a, b }$^*$ }<br>(ix) L = { w \| $n_a$ ( w ) >= 1 ; $n_b$ ( w ) = 2 }<br>(x) L = { w \| $n_a$ ( w ) >= 2 ; $n_b$ ( w ) = 3 }<br>(xi) L = { w \| $n_a$ ( w ) >= 3 ; $n_b$ ( w ) >= 2 }<br>(xii) L = { w \| \| w \| mod 3 >= \| w \| mod 2 }<br>(xiii) L = { w \| \| w \| mod 3 = \| w \| mod 2 where w ∈ ∑$^*$ and ∑ = { a } }<br>(xiv) L = { w \| w ∈( a + b )$^*$ \| $n_a$ ( w ) mod 3 =2 and $n_b$ ( w ) mod 2 = 1<br>(xv) Every block of length Five contains at least two a's. Use Extended Transitions of function to trace a string w = aabba<br>(xvi) The number of a's is divisible by 4 and no of b's divisible by 2<br>(xvii) Set of all strings having odd number of a's and odd number of b's | 2 | 3 |
| 10 | (i)Build a DFA to accept the binary numbers<br><br>(i) divisible by 5<br>(ii) divisible by 3<br><br>(ii)Build a DFA to accept the decimal numbers<br><br>(i) divisible by 5<br>(ii) divisible by 3 | 2 | 3 |
| 11 | Design ε NFA or NFA for the following languages<br><br>(i) abc, abd, aacd on ∑ = { a, b, c, d }<br>(ii) { ab, abc}$^*$ on ∑ = { a, b, c }<br>(iii)L = { abab$^n$ \| n >= 0 } U { aba$^n$ \| n >=0 }<br>(iv)L = { aa$^*$(a + b)} | 2 | 3 |
| 12 | Convert the following NFA to DFA<br><br>a.<br><br>\| δ \| 0 \| 1 \| | 2 | 3 |

| | | |
|---|---|---|
| →$q_0$ | $q_0$ | $q_0$ , $q_1$ |
| $q_1$ | $q_2$ | $q_2$ |
| *$q_2$ | Φ | Φ |

b.



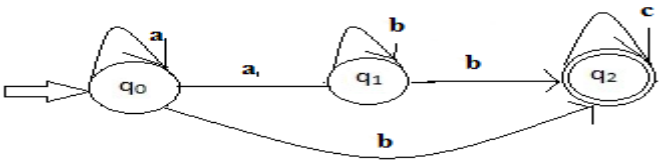

c.



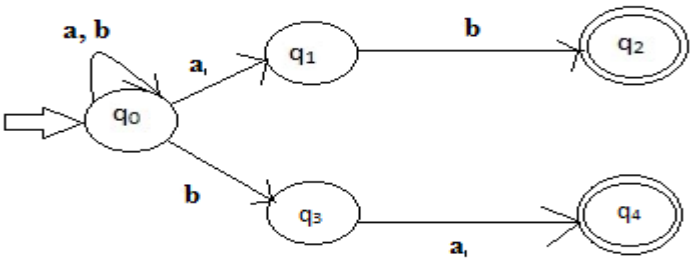

d.

| Δ | A | B |
|---|---|---|
| $q_0$ | $q_0$ , $q_1$ | $q_0$ |
| $q_1$ | - | $q_3$ |
| $q_2$ | - | - |

e.



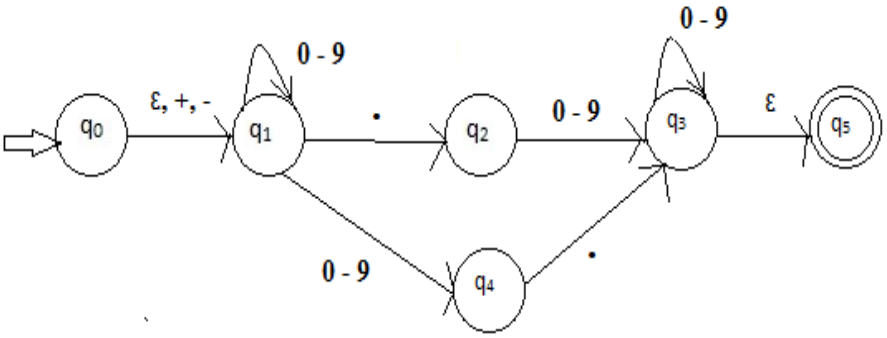

f.



| | | | |
|---|---|---|---|
| 13 | Convert the following NFA to DFA using "Subset construction scheme"<br>a. | 2 | 3 |

| Δ | 0 | 1 |
|---|---|---|
| →p | { p, q } | { p } |
| q | { r } | { r } |
| r | { s } | Φ |
| *s | { s } | { s } |

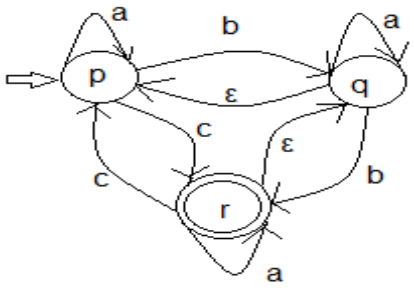
b.




| | | | |
|---|---|---|---|
| 14 | What are the Steps involved in converting ε-NFA to DFA Convert the following ε-NFA to DFA<br>a. | 2 | 3 |

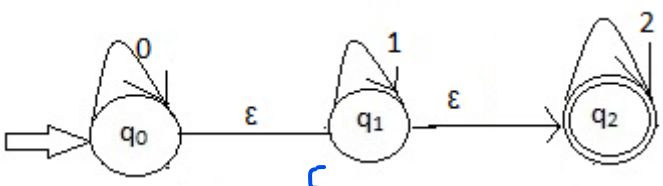
| Δ | E | A | B | C |
|---|---|---|---|---|
| →p | Φ | { p } | { q} | { r } |
| Q | { p } | { q} | { r } | Φ |
| *r | { q} | { r } | Φ | { p } |

b.




c.

d.




| Δ | E | A | B |
|---|---|---|---|
| →p | { r } | { q } | {p, r} |
| Q | Φ | { p } | Φ |
| *r | {p, q} | { r } | { p } |

e.

| Δ | E | A | B | C |
|---|---|---|---|---|
| →$q_0$ | $q_1$ | $q_0$, $q_1$ | $q_2$ | - |
| $q_1$ | $q_2$ | - | $q_1$, $q_2$ | - |
| * $q_2$ | - | - | - | $q_2$ |

**Faculty In-charges**

**Course Coordinator** **Module Coordinator** **IQAC** **Program Coordinator**